

quantms, quantms-rescoring and multiple search engines
enables deep proteome coverage across protein
quantification, immunopeptidomics, and
phosphoproteomics experiments

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Abstract

The exponential growth of public proteomics datasets has surpassed the analytical capacity of traditional desktop tools, particularly for large-scale automated reanalysis. To address this challenge, we present an integrated workflow that combines quantms, a cloud-native pipeline, AlphaPeptDeep and with MS²Rescore, a machine learning-based rescoring tool. This workflow enables deep and scalable reanalysis of massive proteomics datasets. Powered by the Nextflow engine for parallel computing, the pipeline incorporates fragment ion intensity predictions from MS²PIP and retention time predictions from DeepLC, improving peptide-spectrum match reliability through Percolator. We applied this approach to four representative datasets covering label-free quantification, TMT labeling, immunopeptidomics, and phosphoproteomics. Compared to traditional methods, our workflow achieved a 16–22.8% increase in identified spectra, along with the quantification of hundreds of additional proteins and phosphosites. These improvements demonstrate that integrating multiple search engines with machine learning-derived features not only enhances identification sensitivity but also deepens quantitative insights for downstream biological interpretation. Overall, this workflow offers a reproducible and scalable solution for the reanalysis of public proteomics data, advancing FAIR principles by promoting scientific transparency, accessibility, and data reuse.

Keywords: Proteomics, Reanalysis, Workflow, Machine learning.

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1 Introduction

In recent years, the field of proteomics has experienced rapid growth in the availability of publicly accessible datasets, accompanied by a shift toward studies analyzing larger sample cohorts. As of June 2025, over 40,000 datasets have been submitted to ProteomeXchange (PX) repositories, including a substantial increase in large-scale submissions comprising more than 100 instrument files [1]. However, conventional desktop tools such as MaxQuant [2], pFind [3] and PeptideShaker [4] are limited in their capacity to perform automated, large-scale quantitative analyses in cloud or distributed environments, hindering the reanalysis of extensive experiments on standard workstations. Most of these tools (eg. Proteome Discoverer) are commercial or closed-source, although some have begun adopting cloud-based deployment models.

We recently developed quantms [5], an open-source, cloud-based pipeline designed for massively parallel reanalysis of quantitative proteomic datasets. The pipeline is highly modular and flexible, accommodating a wide range of quantitative proteomics approaches including DDA label-free, DDA multiplex (TMT and ITRAQ), and DIA. quantms automatically distributes computations using the Nextflow workflow engine across one or more computing nodes, depending on the number of instrument files and samples [6]. To ensure traceability and reproducibility, the pipeline is built entirely on standardised open file formats and reproducible execution environments, adhering strictly to the FAIR (Findability, Accessibility, Interoperability, and Reusability) principles [7]. quantms has been extensively benchmarked and used in the analysis of large-scale public proteomics datasets [5, 8, 9].

We recently developed quantms, an open-source, cloud-based pipeline designed for massively parallel reanalysis of quantitative proteomics datasets [5]. The pipeline is highly modular and flexible, accommodating a wide range of quantitative proteomics approaches. quantms automatically distributes computations using the Nextflow workflow engine across one or more computing nodes, depending on the number of instrument files and samples [6]. To ensure traceability and reproducibility, the pipeline is built entirely on standard-

ized open file formats [10] [11] and reproducible execution environments such as Docker and Singularity, adhering strictly to the FAIR (Findability, Accessibility, Interoperability, and Reusability) principles [7].

With the adoption of machine learning (ML) in the field of proteomics, various models have been developed to accurately predict peptide behavior in LC-MS, such as MS²PIP [12] and DeepLC [13] for fragment ion intensities and retention time prediction, respectively. Early approaches employed decision trees and single-layer neural networks, while more recent deep learning models such as Prosit [14] achieve significantly improved accuracy for predicting fragment ion intensities and retention times. These highly accurate predictions enable superior matching of experimental data to theoretical expectations and have reinvigorated rescoring strategies in proteomics. MS²Rescore is a modular Python package that generates multiple features assessing the similarity between observed and predicted peptide behavior, such as fragment ion intensities, retention time, and ion mobility [15].

Previously, quantms did not leverage measurable peptide properties such as fragment ion intensities and retention times. To overcome this limitation, we integrated MS²Rescore into quantms and incorporated customized features following Nextflow and nf-core best practices. We demonstrate that the enhanced pipeline supports in-depth analysis of large-scale public proteomics datasets across diverse experimental designs, including label-free quantification (LFQ), tandem mass tag (TMT)-based quantification, immunopeptidomics, and phosphoproteomics studies.

2 Methods

2.1 MS/MS Data and quantms settings

To develop and evaluate the performance of the quantms-integrated MS²Rescore workflow, we selected four publicly available benchmark datasets. Three were obtained from the PRIDE Archive under the identifiers PXD001819, PXD019643, and PXD026824, and one from the

CPTAC data portal under PDC000127. The PXD001819 dataset contains 48 Sigma UPS1 proteins spiked into a background of yeast cell lysate at nine different concentrations: 0.05, 0.125, 0.25, 0.5, 2.5, 5, 12.5, 25, and 50 fmol/ μ L to evaluate quantification performance. We evaluated five different quantms workflow settings: (1) Comet, (2) Comet and MSGF+, (3) Comet with MS²Rescore, (4) Comet and MSGF+ with MS²Rescore, and (5) Comet, MSGF+, and SAGE with MS²Rescore to explore multiple search engines and their integration with MS²Rescore for improved identification and quantification results. All search parameters were the same as described in the previous publication. An FDR filter at 1% was applied at peptide spectrum match (PSM) and protein level at the dataset level. The search results from quantms and MaxQuant at the PSM and protein group levels are provided in Supplemental File S1.

2.2 Rescoring Features and Postprocessing

MS²Rescore computes a broad range of features based on DeepLC-predicted versus observed retention times, as well as MS²PIP-predicted versus observed MS2 fragment ion intensities and then these functions are integrated into quantms-rescoring Python package to compatible with quantms. Some additional features were implemented. For example, Signal-to-noise ratio (SNR) features can be also computed for specific scenarios such as immunopeptidomics. To enhance compatibility and performance, we implements new strategies for model selection, feature export, and model training. For retention time (RT)-related features, retraining of the DeepLC model is supported, allowing the selection of the model with the lowest mean absolute error (MAE) on calibration PSMs. These calibration PSMs are selected from each run at a specified ratio [?]. For MS2 intensity-related features, we perform overfitting tests using the top-scoring 60% of non-decoy PSMs in the calibration set. quantms-rescoring automatically selects the most suitable MS²PIP model for a given batch of PSMs by comparing the correlation scores of predicted and observed intensities across candidate models. Furthermore, it verifies whether the MS²PIP predictions meet predefined correlation and

scoring thresholds. This is achieved by filtering out decoy PSMs, ranking the results by PSM score, and selecting a calibration subset. The method ensures that at least 80% of calibration PSMs exceed the correlation threshold. If none of the models satisfy this criterion, MS²PIP rescoring is skipped unless explicitly forced by the user. This mechanism prevents the use of poorly fitting models and filters out pre-trained models that are incompatible with the data. Additionally, users can specify a list of features for export. All steps are fully parallelized at the run level, significantly reducing runtime for large-scale datasets. Then, the quantms v1.4 workflow calculates a posterior error probability (PEP) for each PSM using Percolator. This is performed under three different feature configurations (1) a baseline model using only search engine-derived features, (2) the baseline model plus MS²Rescore-derived features, and (3) the above combined with SNR features. To merge results from multiple search engines, ConsensusID aggregates PSMs into unified scores based on the calculation of posterior error probabilities (PEP) and chooses the PSM with the highest probability. Final PSM-level q-values are then obtained either directly from Percolator or calculated using OpenMS’ target-decoy approach based on the predicted probabilities. For phosphoproteomics datasets, LuciPHOr2 is employed to assign site-level localization scores and estimate the associated false localization rate using tools from the OpenMS toolkit [?]. All runtime parameters of quantms for different datasets can be found Supplementary file 1.

Significant PSMs were filtered based on q-values, and the FDR was used as a key metric to compare workflows. As shown in Figure 1 1, using consensus scores from two search engines significantly increased identification rates at a fixed FDR threshold. Specifically, combining Comet and MSGF+ improved identified spectra by 17% over Comet alone, and incorporating MS²PIP and DeepLC features through MS²Rescore led to an additional 16% increase. quantms achieved a 28% improvement in the number of PSM identifications compared to MaxQuant. At the quantification level, including MS²Rescore-derived features allowed more low-abundance UPS1 proteins to be quantified (Figure 1B), highlighting the contributions of the integrated workflow. Although MaxQaunt reported more UPS proteins in 250amol,

500amol and 2500amol, the proteins that were only quantified in MaxQuant were all quantified by match between runs (MBR) rather than from MS/MS. The main reason for this is that quantms used different MBR quality control. Therefore, we can observe that quantms achieved lower coefficients of variation (CV) than MaxQuant, considering only proteins that are quantified in at least 50% of the replicates (Figure 1C-D). On the other hand, we also found that the variability of proteins quantified were not significantly reduced or increased after enabling quantms scoring.

3 Results

3.1 quantms with MS²Rescore Enhances Label-Free Identification and Quantification

To systematically evaluate the performance of the quantms-integrated MS²Rescore workflow at both identification and quantification levels, we first analyzed the public benchmark dataset PXD001819. For benchmarking the identifications, five different workflows configurations were designed and compared, including (1) Comet with Percolator, (2) two search engines with Percolator, (3) Comet with MS²Rescore features and Percolator, (4) two search engines with MS²Rescore features and Percolator, and (5) three search engines with MS²Rescore features and Percolator to determine whether features from fragment intensity-based and retention time-based predictors enhanced the identification and quantification process.

To better understand the contribution of individual features, we extracted the top 20 feature weights from Percolator (Supplemental Figure 1). Over half of the top-weighted features were derived from MS²Rescore. Notably, SpecPearsonNorm had a strong positive weight, indicating that a better correlation between predicted and observed intensities improves confidence. Conversely, RtDiffBest had a negative weight, suggesting that large deviations in retention time reduce match quality. Interestingly, peptide length also emerged as a signif-

icant positive feature after MS²Rescore integration, likely because longer peptides generate more fragment ions and thus are more reliably identified.

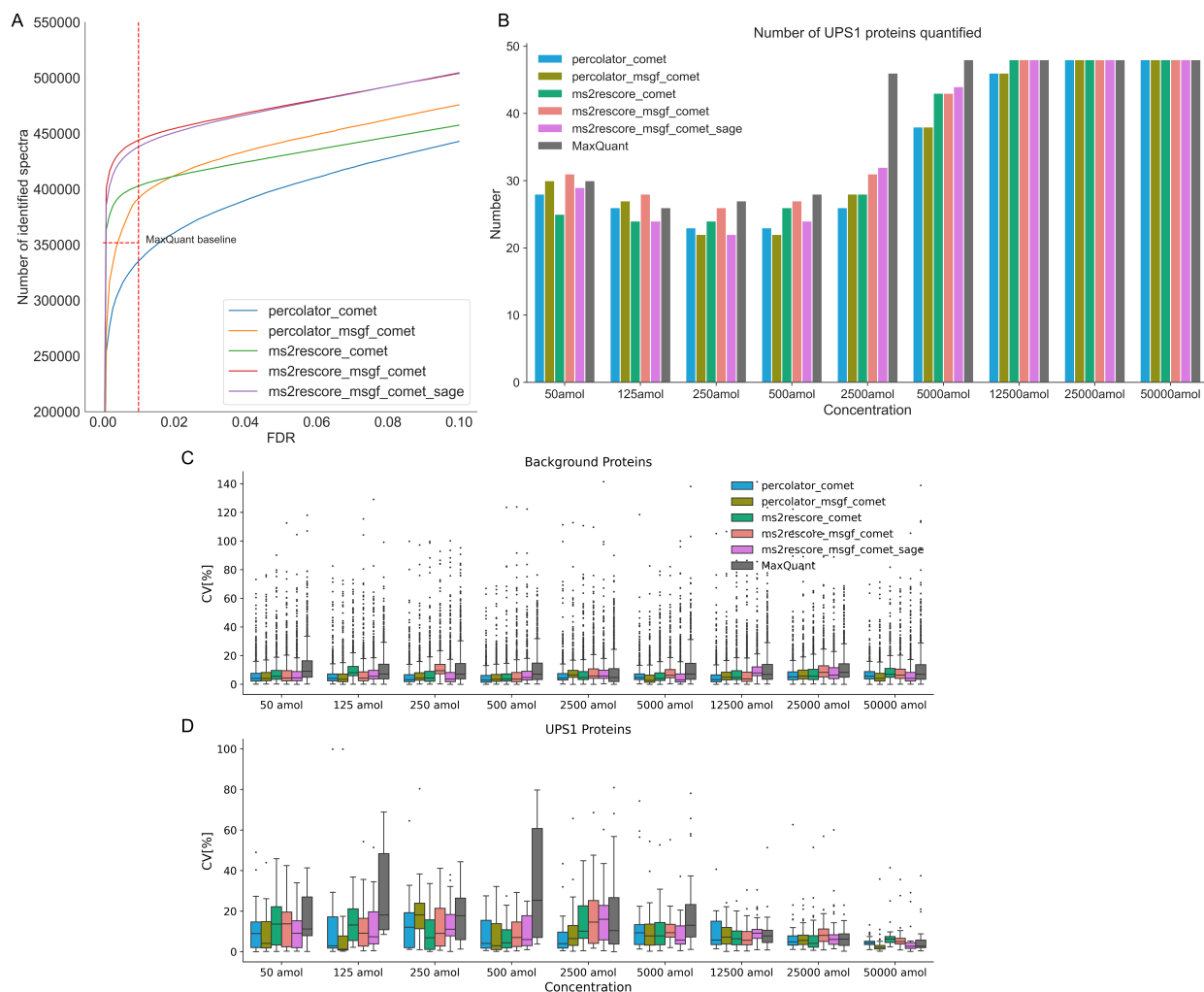


Figure 1: The number of identified spectra as a function of differing FDR levels for different workflow settings. (A) Results for the PXD001819 dataset. (B) Results for the PXD001819 dataset showing protein quantification. Five different workflow settings are shown: the combination of Comet search engine with Percolator rescoring, the combination of Comet with MSGF+, the combination of two search engines with MS2Rescore features, the combination of two search engines with MS2Rescore and SNR features, and the combination of three search engines with MS2Rescore and SNR features. Multiple search engines consensus identification, additional intensity-based and retention time-based features, and SNR features lead to both higher identification rates and high-quality identification. Coefficient of variation of Yeast (C) and UPS1 proteins (D) for the reanalysis of PXD001819 using MaxQuant and quantms with different configurations.

3.2 Improved Identification and Quantification in TMT Experiments via quantms-rescoring

We next applied the workflow to a large-scale TMT-labeled dataset from CPTAC (PDC000127). The integration of multiple search engines and quantms-rescoring led to substantial improvements in both identification and quantification (Figure 2). Compared to Comet with Percolator, multiple search engines consensus identifications with Percolator The PSM identification rate increased by 3.6%, and 921 proteins were newly quantified compared to the workflow without quantms-rescoring (Figure 2A, 2B). Among these, 59 proteins had abundance levels within the top 10% (Figure 2C). To assess the biological impact of rescoring, we conducted differential expression analysis using the MS²Rescore-enhanced workflow. As shown in Figure 2D, 27 newly quantified proteins were significantly differentially expressed. Notably, FOXG1—one of these proteins—has been associated with prognosis in Clear Cell Renal Cell Carcinoma, as reported by Yang et al. [16]. These findings illustrate that quantms incorporating MS²PIP and DeepLC-derived features enhances not only identification sensitivity but also enables deeper biological insights.

We further examined the rescoring contribution of individual features by analyzing the top 20 SVM weights from Percolator (Supplemental Figure S2). For Comet-based results, SpecPearsonNorm, DotProdIonYNorm, and RtDiffBest were highly weighted. The interpretations of these weights were consistent with those from the label-free dataset. Importantly, peptide length again emerged as a key feature. Similar weight distributions were observed for MSGF+ and SAGE results, confirming the generalizability of these feature effects across different search engines. (Supplemental Figure S3)

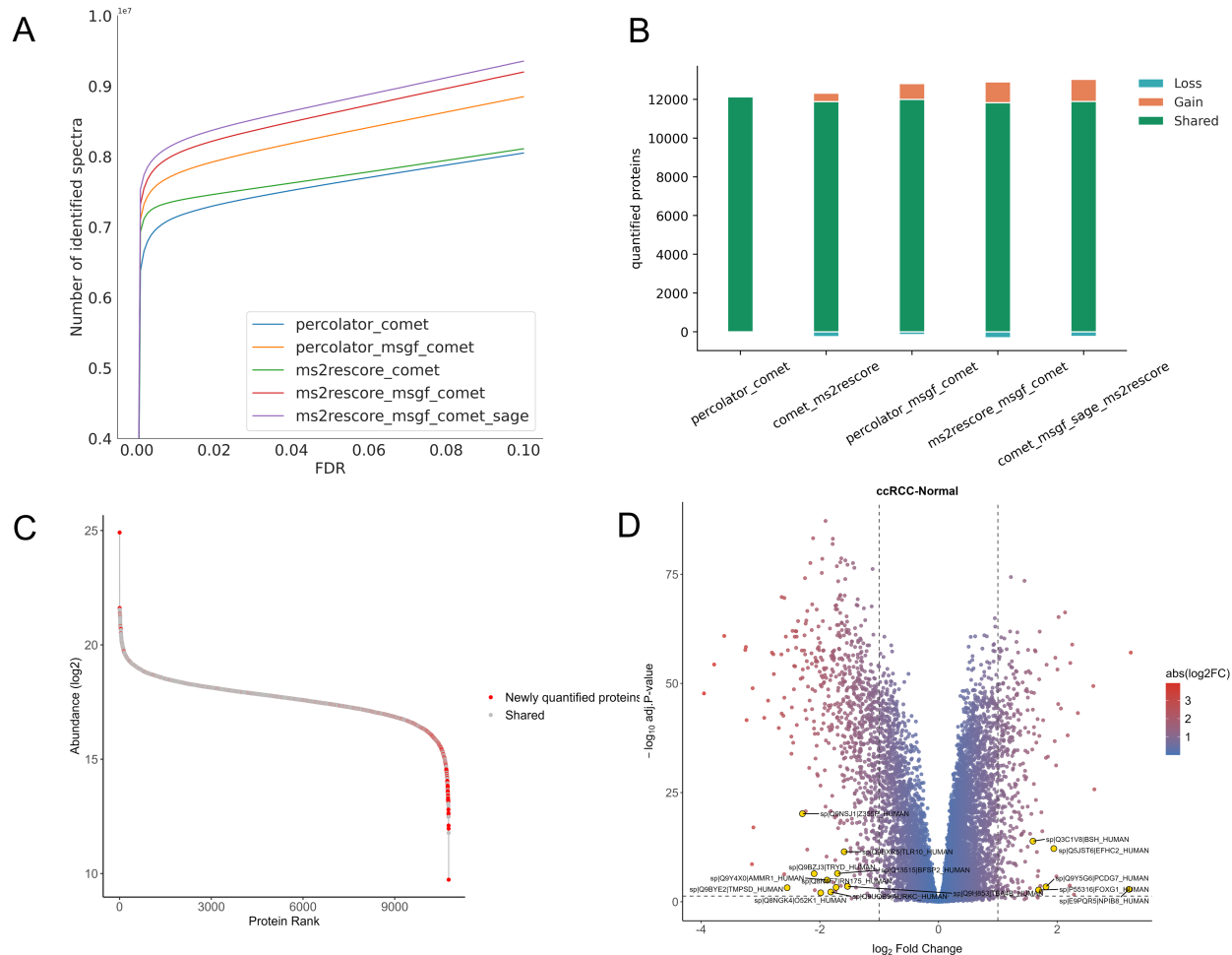


Figure 2: Comparison of identification and quantification results for different workflow settings on PDC000127. (A) The number of identified spectra as a function of differing FDR levels for different workflow settings. (B) The number of quantified proteins. The green part indicates the intersection between a workflow and Percolator-Comet results. (C) The rank of protein abundance from Comet and MSGF+ with MS²Rescore. The red dots represent proteins quantified only in the Comet and MSGF+ with MS²Rescore workflow compared to the Comet and MSGF+ workflow without MS²Rescore. (D) The volcano plot for differential expression analysis from MSstatsTMT. The yellow dots represent proteins quantified only in the Comet and MSGF+ with MS²Rescore workflow compared to the Comet and MSGF+ workflow without MS²Rescore.

3.3 quantms with quantms-rescoring improves HLA Class I immunopeptidome identification

The HLA Ligand Atlas [17] (PXD019643) is a comprehensive map of HLA-I and HLA-II-presented peptides from 30 benign tissues, 51 HLA-I alleles, and 86 HLA-II alleles. To further

assess the performance of the quantms workflow using quantms-rescoring, we analyzed the HLA Class I data of the HLA Ligand Atlas. Five workflow configurations were evaluated: (1) Comet + Percolator, (2) Comet + quantms-rescoring, (3) Comet + MSGF+ + Percolator, (4) Comet + MSGF+ + quantms-rescoring features, and (5) Comet + MSGF+ + quantms-rescoring + SNR features. These configurations were compared based on both the total number of peptide-spectrum matches (PSMs) and the number of unique peptide sequences identified. Across all comparisons, the use of multiple search engines combined with quantms-rescoring features significantly enhanced identification performance. At both 1% and 0.1% FDR thresholds, integrating MSGF+ with Comet increased the number of identified spectra by 11.7% compared to using Comet alone. Adding quantms-rescoring features on top of the multi-engine workflow yielded a further 22.8% improvement, as shown in Figure 3.

To better understand the impact of quantms-rescoring-derived features on rescoring, we visualized the distributions of decoy PSMs, rejected target PSMs, and accepted target PSMs using the Pearson correlation coefficient (PCC) and retention time error (Figure 3C, D). The accepted target PSMs were clearly separated from both decoy and rejected targets based solely on PCC values, and exhibited consistently low retention time deviations. The top 20 feature weights from Percolator, displayed in Supplemental Figure S4. The features of the MSGF+ search engine carry significant weight, such as SpecEValue, which indicates that when only MSGF+ is used, the gain of quantms-rescoring is limited. However, features from quantms-rescoring occupy a larger weight for Comet search engine, including 9 MS²PIP and 2 DeepLC features. This further illustrates that using consensus identifications by multiple search engines helps to coordinate the differences between different search engines.

In addition, A peptide-level FDR threshold of 1% was applied to all MS Runs. To gauge the biological reasonability of these peptides, we searched for sequence motifs of both the phosphorylated and non-phosphorylated peptides. This revealed the expected HLA peptide motifs, dominated by the anchor residues for their cognate major histocompatibility complex proteins.

These patterns were observed across both individual search engine support and multi-engine consensus, further confirming that the integration of intensity and RT-based features enhances PSM discrimination. Overall, these results demonstrate the robustness of the quantms workflow in immunopeptidomics, highlighting its ability to leverage machine learning-derived features to maximize identification accuracy through multi-engine consensus rescoring.

3.4 Enhanced Phosphopeptide Identification and Localization Using quantms-rescoring

In addition, we investigated the performance of our quantms workflow with quantms-rescoring on post-translational modification experiments (PXD026824). For phosphoproteomics analyses, different workflow settings were evaluated: (1) Comet + Percolator, (2) Comet + quantms-rescoring, (3) Comet + MSGF+ + Percolator, and (4) Comet + MSGF+ + quantms-rescoring features. As shown in Figure 4, combining search engine consensus results with quantms-rescoring features substantially improves Percolator’s ability to discriminate between true and false PSMs. This is evident from the leftward shift in score distributions compared to the Comet-only workflow, indicating improved stringency (Figure 4A). Specifically, incorporating quantms-rescoring features into the two-search-engine consensus increased the number of identified spectra by 24% compared to consensus identification alone. Previous research suggest that phosphorylated peptides may produce -98 Da phosphoric acid neutral losses. Therefore, we further evaluate the impact of considering neutral loss on re-scoring performance when calculating features. quantms-rescoring with modloss ions increased the number of identified spectra by 3% compared to without modloss ions. The top 20 feature weights from Percolator, displayed in Supplemental Figure S5, highlight the consistent importance of features such as IonYNorm and DotProdIonYNorm, similar to previous datasets.

Given the importance of false localization rate (FLR) control in phosphoproteomics, we

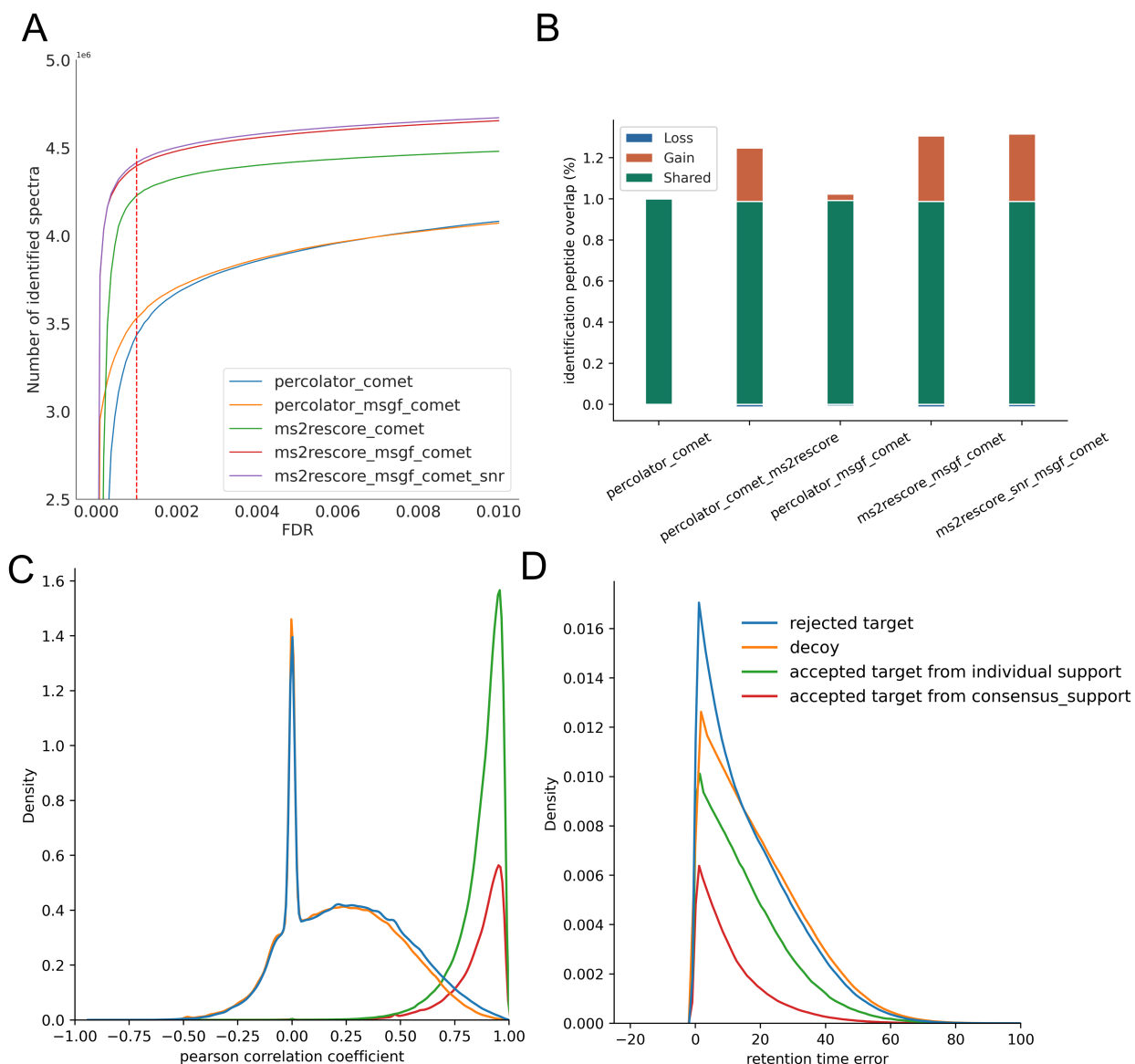


Figure 3: Comparison of immunopeptides identification results for different workflow settings on PXD019643. (A) Line plot showing the number of spectra identified for four workflow settings at different PSM FDR levels. (B) Percentage of unique identified peptides using different workflow settings. Density plots showing the distribution of the smallest retention time error between observed and predicted retention time (C) and the Pearson correlation between observed and predicted peak intensities (D) for each PSM split into decoys (red), rejected targets with q-value >0.01 (blue), and accepted targets from individual support and consensus support with q-value <0.01 (green). Note that the rejected target distribution coincides with the decoy distribution. The pronounced spike at Pearson correlation = 0 reflects spectrum comparisons in which all predicted or observed peaks have zero intensity, causing a division-by-zero situation that leaves the Pearson correlation undefined.

also assessed the impact of different workflows on phosphopeptide identification at varying false localization rate (FLR) thresholds (Figure 4B, C). After filtering with PSM FDR 0.01 and local FLR 0.01, the consensus workflow with quantms-rescoring features identified 20% and 23% more phosphorylated peptides than its counterpart without quantms-rescoring and single search engine. Notably, after filtering with PSM FDR 0.01, FLR 0.01 and protein FDR 0.01, 1,809 phosphorylated peptides were uniquely quantified by the quantms-rescoring enabled workflow compared to without quantms-rescoring. When considering -98 Da neutral losses, 2,191 phosphorylated peptides were uniquely quantified compared to without quantms-rescoring. At the phosphosite level, workflow with quantms-rescoring also uncovered 842 novel protein phosphorylation sites not reported in the other workflow configuration without quantms-rescoring (Figure 4D), and an additional 495 novel protein phosphorylation sites are reported when considering -98 Da neutral losses. Collectively, these results underscore the value of quantms-rescoring in enhancing both identification sensitivity and site-level localization in phosphoproteomics, establishing quantms as a robust solution for large-scale PTM data analysis.

4 Discussion

Advancements in deep learning-based tools have significantly improved the sensitivity of peptide-spectrum match (PSM) rescoring, yet their integration into streamlined, reproducible, and quantitative proteomics workflows has remained limited, especially for large-scale public data analysis. In this study, we demonstrate the systematic incorporation of MS²Rescore into the open-source quantms pipeline and evaluate its impact across various experimental settings, including label-free quantification (LFQ), tandem mass tag (TMT)-based quantification, immunopeptidomics, and phosphoproteomics. Our results show that enhanced feature sets derived from MS²PIP and DeepLC not only increase the identification rates but also improve the quantification depth. Importantly, we provide evidence that

the improved identification process via rescoring propagates downstream into quantification, leading to increased numbers of proteins with reliable abundance estimates, as well as enhanced detection of differentially expressed proteins, and enhanced localization of PTMs.

To demonstrate the applicability and advantages of quantms integrated with MS²Rescore in quantitative proteomics, we performed a series of reanalyses using publicly available large-scale datasets with well-established benchmarks. In label-free experiments, MS²Rescore-enhanced workflows achieved significant increases in PSM and peptide identification rates while maintaining or improving quantification reproducibility across replicates. For TMT experiments, the integration of MS²Rescore into quantms increased the number of quantifiable proteins. In more challenging applications, such as immunopeptidomics and phosphoproteomics, the deep learning-based rescoring pipeline consistently yielded higher identification rates and improved sensitivity. These improvements will contribute to meaningful biological insights, such as phosphosite discovery and the identification of differentially expressed proteins with potential clinical relevance. In addition, we counted the run times with different configurations and the added sensitivity from rescoring does add 15%-50% of run time, as shown as Supplemental Table S1. This can be alleviated by increasing the number of quantms-rescoring parallel processes.

Overall, the results highlight the synergistic benefits of integrating machine learning-based features within end-to-end, cloud-based proteomics workflows. By embedding MS²Rescore into the quantms pipeline, we provide a practical and accessible solution for the community to leverage cutting-edge spectral prediction and retention time modeling without the need for complex manual configuration. This integrated approach enables reproducible reanalysis of large public datasets, increases identification confidence, and strengthens quantitative conclusions across diverse experimental designs. Future work will focus on expanding this approach to other types of post-translational modifications and exploring the integration of additional machine learning tools to further enhance proteomics data analysis capabilities.

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Table S1: Runtime, memory, and CPU consumption for different configurations. The memory and CPU values represent the maximum resources required for a single job. The quantms workflow was run in a 64 cluster queue.

Datasets	Configuration	Runtime	Memory	CPU
PDC000127	Comet&MSGF+&MS ² Rescore	9h39m	30G	7
PDC000127	Comet&MSGF+	6h28m	8G	4
PDC000127	Comet	5h10m	8G	4
PXD019643	Comet&MSGF+&MS ² Rescore	15h57m	20G	8
PXD019643	Comet&MSGF+	13h09m	5.5G	4
PXD019643	Comet	2h05m	5.5G	4
PXD026824	Comet&MSGF+&MS ² Rescore	2h05m	66G	8
PXD026824	Comet&MSGF+	1h17m	48.4G	4
PXD026824	Comet	24m	48.4G	4

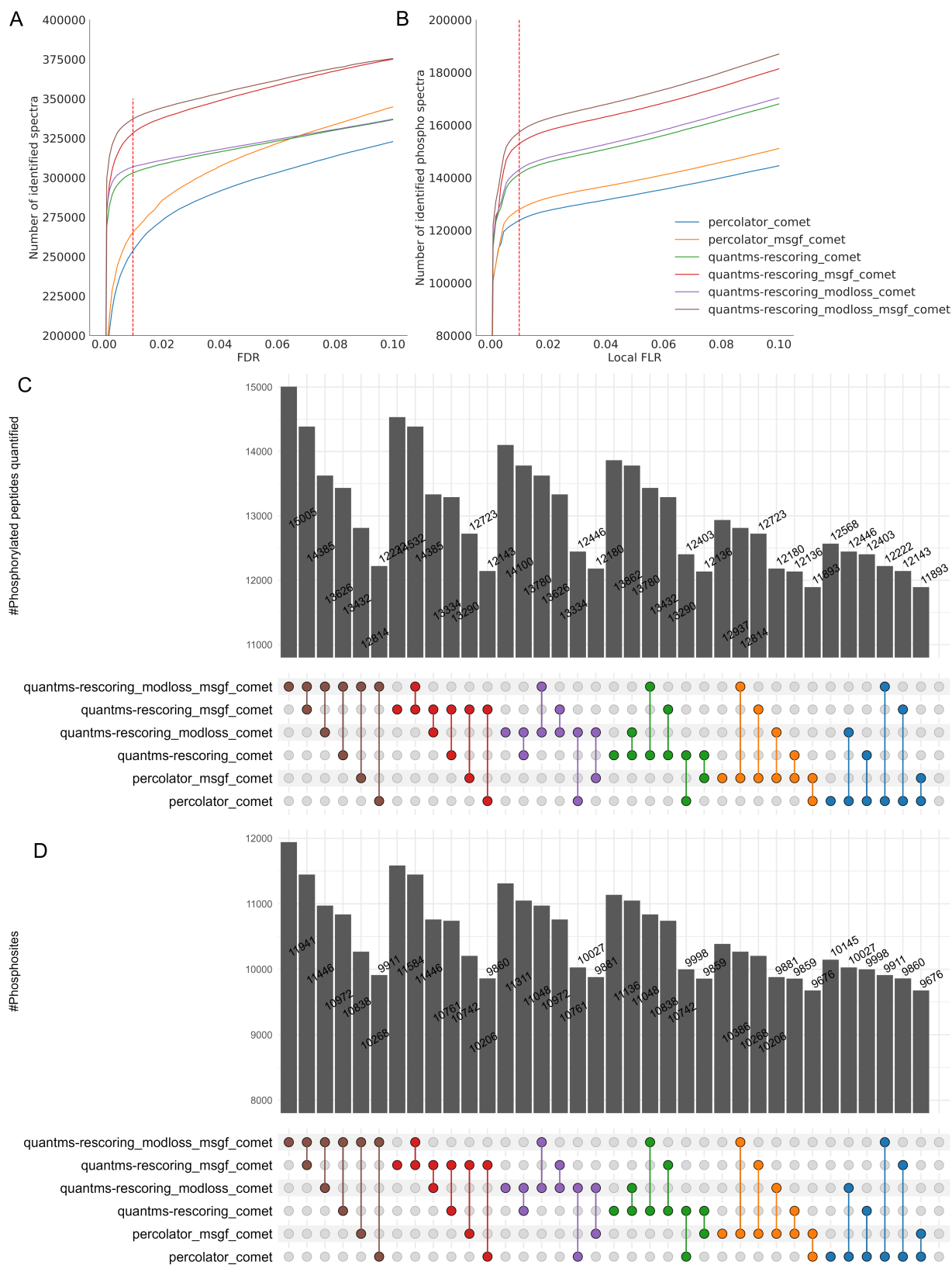


Figure 4: Comparison of phosphorylated peptide identification results for different workflow settings on PXD026824. (A) Line plot showing the number of spectra identified for four workflow settings at different PSM FDR levels. (B) Line plot showing the number of phosphorylated spectra identified for four workflow settings at different local false localization rate levels after an FDR of less than 0.01 at the PSM level. (C) Venn diagram of peptides quantified for four settings. (D) Venn diagram of protein phosphosites at protein FDR 0.01

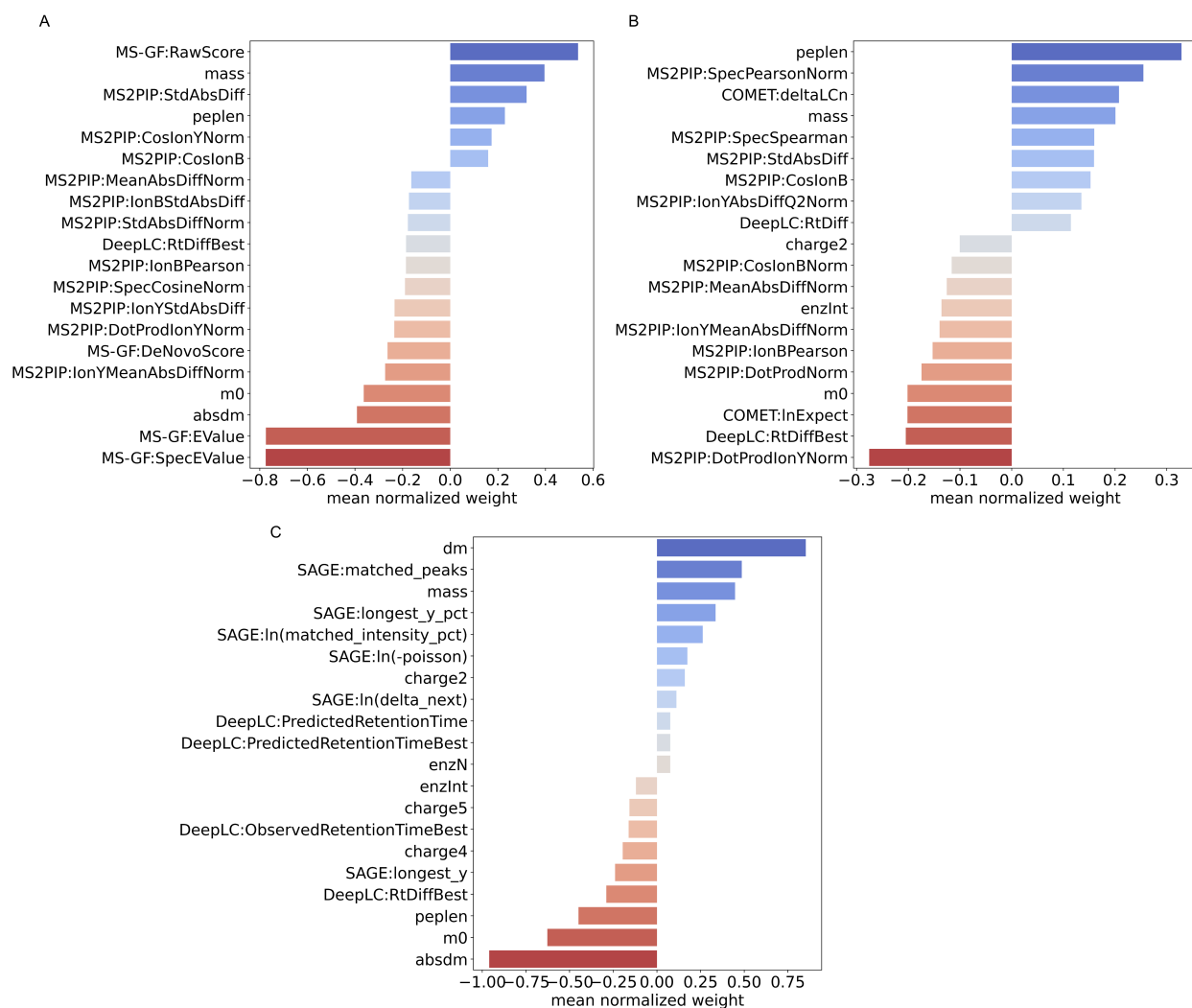


Figure S1: The weights that Percolator assigns to different features of the search engine feature vector in MSGF+ (A), Comet (B), and SAGE (C) combined with Percolator settings from PXD001819. The more different from zero a weight is (both positive and negative), the more important that feature is in the final Percolator classification.

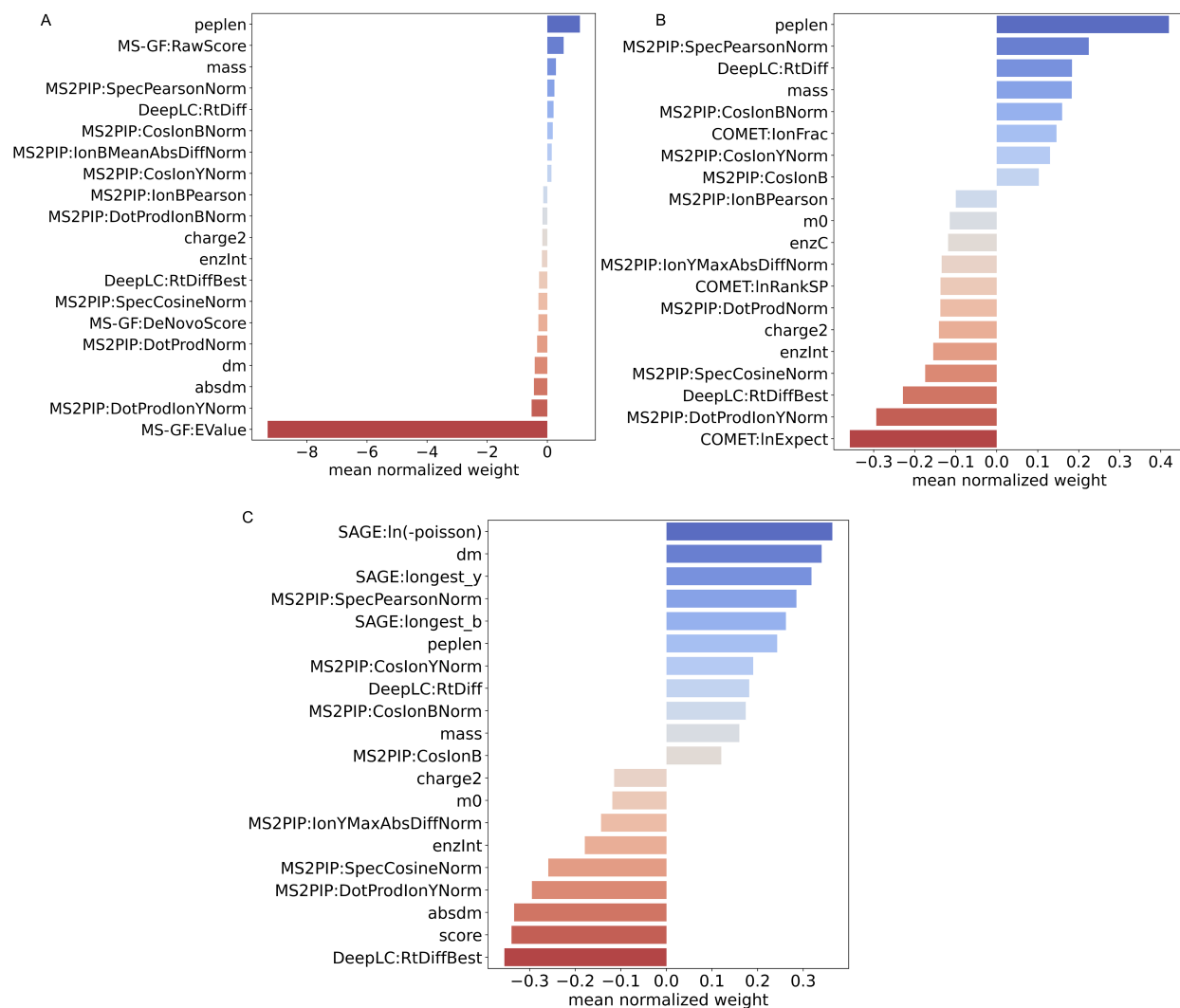


Figure S2: The top 20 normalized weights from Percolator for MSGF+ (A), Comet (B), and SAGE (C) identification results from PDC000125.

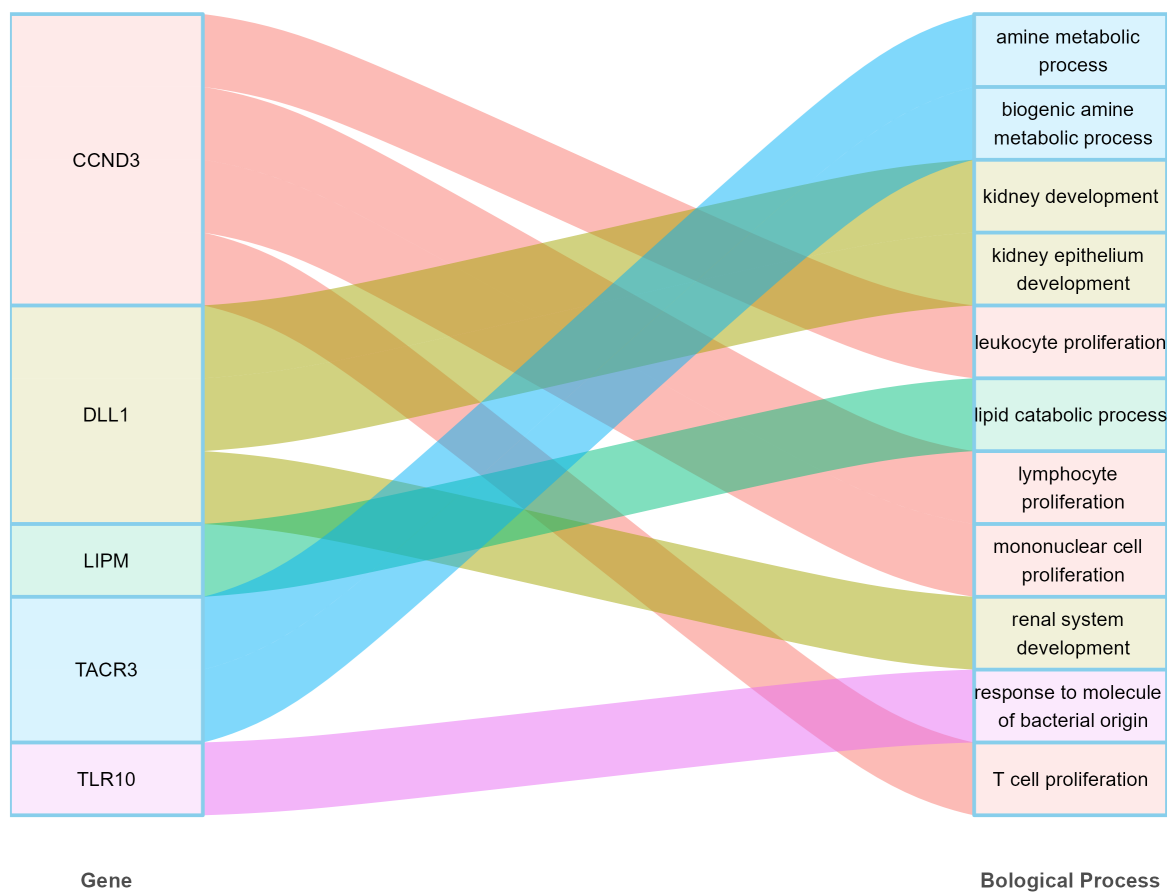


Figure S3: The Go enrichment analysis for MSGF+&Comet with quantms-rescoring. The Sankey diagram shows biological process term of the genes are uniquely reported by MSGF+&Comet with quantms-rescoring compared without quantms-rescoring.

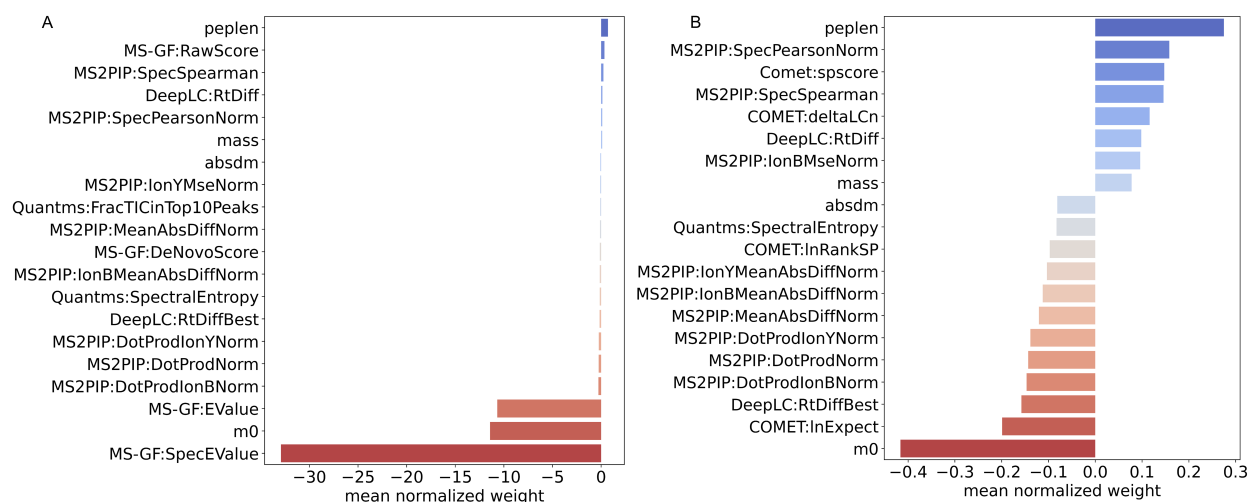


Figure S4: The top 20 normalized weights from Percolator for MSGF+ (A) and Comet (B) identification results in the PXD019643 dataset.

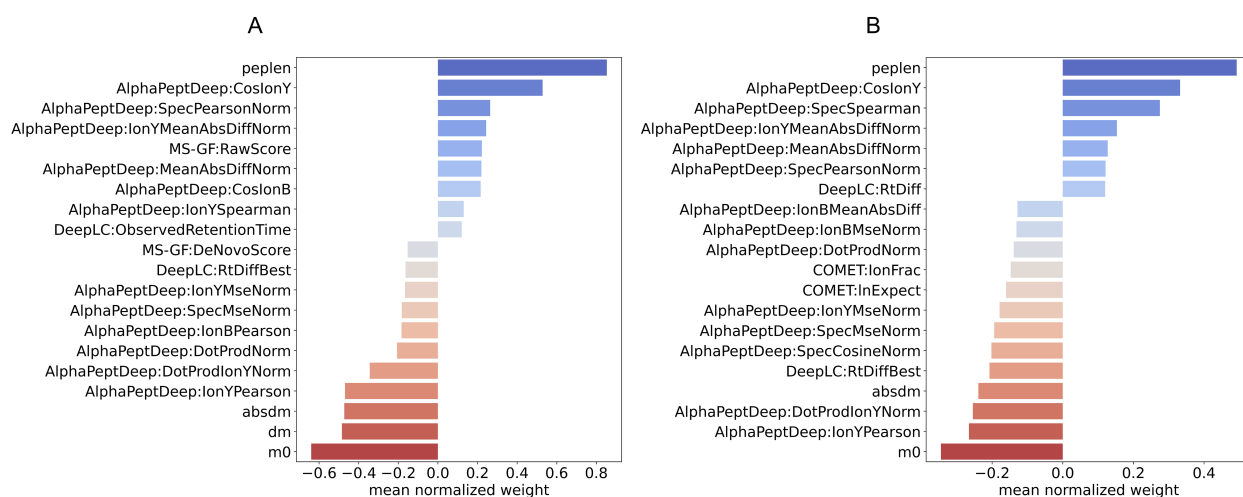


Figure S5: The top 20 normalized weights from Percolator for MSGF+ (A) and Comet (B) identification results in the PXD026824 dataset.